MIL-C-13125C(MR)
18 January 1983
SUPERSEDING
MIL-C-13125B(MR)
9 December 1968

MILITARY SPECIFICATION

COMPOUND: MOUNTING FOR OPTICAL LENS BLANKS

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. Mounting compound covered by this specification shall be a resin and filler mixture put up in bars (see 3.3) for mounting optical lens blanks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-639 - Brass, Naval, Flat Products (Plate, Bar, Sheet, and Strip)
PPP-B-585 - Box, Shipping, Fiberboard, Wood, Wirebound
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Fiberboard
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SSS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 5 - Penetration of Bituminous Materials, Test for

D 36 - Softening Point of Bitumen (Ring-and-Ball Apparatus), Test for

D 127 - Drop Melting Point of Petroleum Wax Including Petrolatum, Test for

D 792 - Specific Gravity And Density of Plastics by Displacement, Test for

D 974 - Neutralization Number by Color-Indicator Titration, Test for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The mounting compound shall be a mixture of resin and filler furnished in bars (see 3.3). The compound shall be solid at room temperature and shall have sufficient adhesion to hold optical lens blanks rigidly to a metal tool during the lens grinding and polishing operation.

3.2 Preproduction. Unless otherwise specified (see 6.2), mounting compound furnished under this specification shall be a product that has been tested and has passed the preproduction tests (see 4.5(a)).

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3.3 Shape and size. Unless otherwise specified (see 6.2), mounting compound shall be furnished in the shape of rectangular bars 3/4 ± 1/8 inch, by 5/8 ± 1/8 inch, by 11 ± 1/4 inches (see 4.4).

3.4 Physical properties. The mounting compound shall conform to the physical properties shown in table I.

Table I. Physical properties

Test	Requirement	Test paragraph
Specific gravity	1.50 to 1.70	4.6.1
Penetration at 54.4°C	15 to 20 mm	4.6.2
Softening point	90.6° ± 5.6°C	4.6.3
Melting point	93.3° ± 5.6°C	4.6.4
Pot life	(see 3.4.1)	4.6.5
Odor	not unpleasant at 98.9°C	4.6.6
Acid number	20 to 65	4.6.7
Low temperature	(see 3.4.2)	4.6.8
Adhesion	200 to 360 lbs/sq. inch	4.6.9
Water immersion	(see 3.4.3)	4.6.10

3.4.1 Pot life. Mounting compound tested as indicted in 4.6.5 shall meet the requirements for adhesion and penetration.

3.4.2 Low temperature. The mounting compound at -13°F (-25°C) minimum shall be easily removed from glass when tested as specified in 4.6.8.

3.4.3 Water immersion. The mounting compound shall comply with the adhesion requirements of table I (200 to 360 pounds per square inch) after 24 hours water immersion when tested as specified in 4.6.9.

3.5 Workmanship. The mounting compound shall be uniform in quality and free from coarse matter and unbroken lumps of filling material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot. A lot of material shall consist of all material made by the same process from the same composition by one manufacturer and submitted for inspection at one time.

4.3 Sampling.

4.3.1 For examination. Unless otherwise specified, a random sample of filled containers shall be selected for examination in accordance with level 1 to MIL-STD-105.

4.3.2 For tests. Unless otherwise specified, samples for testing shall be selected from each batch or shipment in accordance with level S-1 of MIL-STD-105.

4.4 Examination. Sample units selected in accordance with 4.3.1 shall be examined for the defects and at the acceptable quality level shown in table II.

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Table II. Classification of defects

	AQL percent	Classification of defect	Defect	Method of inspection
		Critical	None defined	
Mounting compound (see 4.3.1, 3.3 and 3.5)	1.5	Major 101	Not in bars	Visual
		Major 102	Wrong dimensions on bars	Measure
		Major 103	Not uniform in quality	Visual
		Major 104	Not free from coarse matter	Visual
		Major 105	Not free from unbroken lumps of filling material	Visual
Inner cartons (see 4.3.1 and 5.1)	1.5	Major 106	Wrong type	Visual
		Major 107	Wrong size	Visual
		Major 108	Improperly closed	Visual
Box open (see 4.3.1 and 5.2)	1.5	Major 109	Improper type	Visual
		Major 110	Improper size	Visual
Box closed (see 4.3.1, 5.2 and 5.3)	1.5	Major 111	Lack of, or improper strapping	Visual
		Major 112	Improperly closed	Visual
		Major 113	Improperly marked	Visual
		Major 114	Excessive weight	Approved scale 1/

1/ Approved by procuring agency

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4.5 Classification of tests. The testing and inspection under this specification are classified as follows:

(a) Preproduction tests. Prior to production, preproduction tests shall be conducted to determine compliance with all requirements of this specification (see 6.2).

(b) Acceptance tests. Acceptance tests shall consist of all tests of this specification. Failure to comply with any of the test requirements of this specification shall be cause for rejection of the represented material.

4.6 Tests.

4.6.1 Specific gravity. Specific gravity shall be determined by ASTM D 792 except that the test temperature shall be 25° ± 1.1°C (77° ± 2°F) instead of 23°C.

4.6.2 Penetration. Penetration shall be determined in accordance with ASTM D 5 using standard needles with a 50 gram load. The reading on indicator shall be as listed in table I.

4.6.3 Softening point. The softening point shall be determined in accordance with ASTM D 36. The thermometer used in this determination shall have a range up to 392°F (200°C).

4.6.4 Melting point. The melting point shall be determined in accordance with ASTM D 127 except that a thermometer having a range up to 392°F (200°C) may be used.

4.6.5 Pot life. Melt the mounting compound and maintain the temperature at 210°F ± 3°F (98.9° ± 1.7°C), for 8 hours. At the end of 8 hours, repeat the adhesion test as indicated in 4.6.9 and penetration as indicated in 4.6.2.

4.6.6 Odor. During the pot life test, when the mounting compound is heated at 210°F (98.9°C), olfactory examinations shall be made by 3 to 4 individuals for unpleasant odor. Two determinations at least an hour apart shall be made.

4.6.7 Acid number. The acid number shall be determined in accordance with ASTM D 974.

4.6.8 Low temperature. Two brass plates, 4 by 6 by 1/4 inches, and 12 optical lens blanks 1-1/2 inches in diameter by 1/4 inch thick shall be cleaned as described in 4.6.9.1.1 and 4.6.9.1.2. After drying, coat plates on one side with the compound as prepared in 4.6.5 to a thickness of at least 1/4 inch. Then place 6 lens blanks on each plate and allow to stay at room temperature for not less than one hour. After that, expose the plates and lens blanks to a temperature of -13°F (-25°C) for not less than two hours. After 2 hours cooling, remove the plates and tap them on the table. Observe for requirements in accordance with 3.4.2.

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4.6.9 Adhesion.

4.6.9.1 Preparation of brass strips and glass (microslides).

4.6.9.1.1 Cleaning strips. Strips, half-hard (conforming to QQ-B-639), 1 inch by 4 inches (minimum thickness-1/16 inch) shall be cleaned in solvent (toluene or benzene), then with alkaline cleaner (2 percent sodium metasilicate plus 0.5 to 1.0 Nacconol NR) and rinsed with hot tap water. Strips then shall be pickled for 1 to 3 minutes at temperature 160° to 180°F (71.1° to 82.2°C). (Pickling solution - 85 g sodium dichromate, 245 cc concentrated sulfuric/acid and add enough water to make 2 liters of solution). After pickling, strips shall be rinsed with hot distilled water and air dried.

4.6.9.1.2 Cleaning glass. Put glass slides into the cleaning solution and keep in for 5 minutes. After that the slides shall be given thorough hot tap water rinse and final rinse shall be with hot distilled water. (Prepare the cleaning solution as follows: To 20 grams of sodium dichromate, add enough water to make a thick paste, then add 300 cc of concentrated sulfuric acid).

4.6.9.2 Assembling the strips and glass. Place brass strips, as specified in 4.6.9.1.1, on hot plate and heat them to not more than 82°F (27.8°C) above the melting point of the mounting compound. At this temperature, rub the compound over the one end of strip (1 square inch) to produce very thin layer. Remove coated strips and allow to cool to room temperature. Then place a clean glass (micro-slide 1 inch square) between two coated strips to form a lap-shear tensile specimen with the bond or overlap in the center of the specimen. Secure the bond with scotch tape. The assembly shall be placed on hot plate for 2 minutes with a 5 pound weight over the bonded area. Remove specimen from hot plate and cool to room temperature. Excess compound on the slides or strips shall be scraped off. At least 5 assemblies shall be prepared.

4.6.9.3 Tensile strength test. The free ends of an assembly, as specified in 4.6.9.2, shall be fastened in the jaws of a tensile testing machine and pulled at a rate of approximately 0.05 inch per minute. The force required to shear the specimen shall be recorded.

4.6.10 Water immersion. Samples (assemblies) prepared as specified in 4.6.9.2 shall be soaked in water at room temperature 73.4° ± 2.7°F (23° ± 1°C) for 24 hours. Tests for adhesion shall be made in accordance with 4.6.9.3.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. Unless otherwise specified, mounting compound shall be furnished with 4 bars in a fiberboard box conforming to PPP-B-636 and 6 of these boxes in an intermediary fiberboard box also conforming to PPP-B-636. When tape is used, closure shall be accomplished using 2 strips conforming to PPP-T-76. Tape shall be 2 inches wide and shall be applied to top and bottom seams. If the bottom of the container is closed with staples or adhesive (or a stype OPF container is used) only a single piece on the top seam is required. Tape need not extend onto the end panel of the container.

5.1.2 Level C. Unless otherwise specified, the mounting compound shall be packaged in accordance with standard commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified by the procuring activity, material packaged as specified in 5.1 shall be packed in exterior cases of a type as follows:

PPP-B-585, style 2, 2A or 3, type 3: Boxes, Wood, Wirebound.
PPP-B-591, overseas type: Boxes, Fiberboard, Wood-cleated.
PPP-B-601, overseas type: Boxes, Wood, Cleated-Plywood.
PPP-B-621 - overseas type; style 4: Boxes, Wood, Nailed and Lock-Corner
PPP-B-636, - Class Weather-Resistant.

The gross weight of wood-cleated boxes shall not exceed 200 pounds and of fiberboard boxes the limitations of the box specifications. As far as practical, containers shall be of uniform shape and size, be of a minimum cube and tare consistent with the protection required and contain identical quantities. Containers shall be closed and strapped in accordance with applicable container specification or appendix thereto.

5.2.2 Level B. Material packaged as specified in 5.1 shall be packed in containers conforming to the specifications listed in 5.2.1 except that the containers shall be domestic type. Weight limitations for wood-cleated and wirebound containers shall not exceed 200 pounds. The weight limitation of the domestic-type fiberboard containers shall be as specified within the limitation of the box specification. As far as practical, containers shall be of uniform shape and size, be of a minimum cube and tare consistent with the protection required, and contain identical quantities. Closure of the boxes shall be as required by the appendix of the specification for domestic or level B packing.

5.2.3 Level C. Packing shall be in accordance with commercial practice adequate to insure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 Marking. Each carton, box, or shipping container shall be plainly marked with the following:

Compound; Mounting, for Optical Lens Blanks.
Specification MIL-C-13125C(MR).
Name of Manufacturer.
Contract or Purchase Order.
Date packaged (month and year).

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In addition, shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Mounting compound covered by this specification is intended for use in mounting lenses on metal tools in order to further process lenses on a polishing machine.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether preproduction sample is required (see 3.2).
(c) If required, where preproduction sample should be sent.
(d) Size of bar, if other than that specified (see 3.3).
(e) Level of packaging and packing required (see 5.1 and 5.2).

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